

Quantitative Aspects of Chemical Change — Complete Notes

Grade 11

Physical Sciences

CAPS

What you'll learn

- Use the mole concept and Avogadro's number to count particles.
- Convert between mass, moles and number of particles.
- Use molar mass and concentration in calculations.
- Balance equations and solve stoichiometry problems (mass–mass and solution).
- Identify the limiting reagent and calculate percentage yield and purity.
- Determine empirical and molecular formulae.

The mole and Avogadro's number

A mole (mol) is the amount of substance that contains the same number of particles as there are atoms in 12 g of carbon-12. That number is Avogadro's number:

$$N_A = 6,022 \times 10^{23} \text{ particles} \cdot \text{mol}^{-1}$$

The particles can be atoms, molecules, ions or formula units. The mole is simply a chemist's counting unit, just as a dozen means 12.

Key formulas

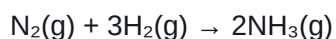
Formula	Meaning	Units
$n = m / M$	amount from mass	mol (m in g, M in $\text{g} \cdot \text{mol}^{-1}$)
$N = n N_A$	number of particles	particles
$n = c V$	amount from concentration	mol (c in $\text{mol} \cdot \text{dm}^{-3}$, V in dm^3)
$c = n / V$	concentration	$\text{mol} \cdot \text{dm}^{-3}$
$n = V / V_m$	amount of gas at STP	mol ($V_m = 22,4 \text{ dm}^3 \cdot \text{mol}^{-1}$)

Molar mass M is the mass of one mole of a substance, in $\text{g} \cdot \text{mol}^{-1}$. It is numerically equal to the relative formula mass. Example: $M(\text{NaCl}) = 23 + 35,5 = 58,5 \text{ g} \cdot \text{mol}^{-1}$.

Note: $1 \text{ dm}^3 = 1 \text{ litre} = 1\,000 \text{ cm}^3$. To convert cm^3 to dm^3 , divide by 1 000.

Stoichiometry

A balanced equation gives the mole ratio in which substances react. For example:



means 1 mol N_2 reacts with 3 mol H_2 to form 2 mol NH_3 . The coefficients tell you the ratio, not the masses.

Steps for a stoichiometry calculation:

1. Write and balance the equation.

2. Convert the given quantity to moles.
3. Use the mole ratio from the balanced equation.
4. Convert moles of the required substance to the quantity asked for (mass, volume, concentration).

Limiting reagent

When reactants are not present in the exact ratio of the balanced equation, one runs out first — the limiting reagent. It determines the maximum amount of product formed. The other reactant is said to be in excess.

To find it: calculate the moles of product that each reactant could form; the smallest value comes from the limiting reagent.

Percentage yield and purity

Percentage yield = (actual yield / theoretical yield) $\times$ 100

Percentage purity = (mass of pure substance / mass of sample) $\times$ 100

The theoretical yield is calculated from the limiting reagent. The actual yield is what you really obtain in the lab, and it is always less than or equal to the theoretical yield.

Empirical and molecular formula

- Empirical formula = the simplest whole-number ratio of atoms in a compound.
- Molecular formula = the actual number of atoms; it is a whole-number multiple of the empirical formula.

From percentage composition: divide each percentage by the element's atomic mass, then divide all the answers by the smallest one, and round to whole numbers. Then $M_{\text{molecular}} / M_{\text{empirical}}$ gives the multiplier.

Worked example 1 — mass, moles and particles

Calculate the mass of 0,50 mol of sodium carbonate, Na_2CO_3 , and the number of formula units present.

Step 1 — Molar mass:

$$M(\text{Na}_2\text{CO}_3) = 2(23) + 12 + 3(16) = 46 + 12 + 48 = 106 \text{ g}\cdot\text{mol}^{-1}$$

Step 2 — Mass:

$$m = n M = 0,50 \times 106 = 53 \text{ g}$$

Step 3 — Number of formula units:

$$N = n N_A = 0,50 \times 6,022 \times 10^{23} = 3,011 \times 10^{23} \approx 3,0 \times 10^{23} \text{ formula units}$$

Worked example 2 — solution stoichiometry

25,0 cm³ of hydrochloric acid of concentration 0,100 mol·dm⁻³ exactly neutralises a sodium hydroxide solution. Calculate the mass of NaOH that reacted.

Equation: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (mole ratio 1 : 1).

Step 1 — Convert the volume to dm³:

$$V = 25,0 \text{ cm}^3 = 0,0250 \text{ dm}^3$$

Step 2 — Moles of HCl:

$$n = c V = 0,100 \times 0,0250 = 2,50 \times 10^{-3} \text{ mol}$$

Step 3 — Mole ratio HCl : NaOH = 1 : 1, so:

$$n(\text{NaOH}) = 2,50 \times 10^{-3} \text{ mol}$$

Step 4 — Molar mass of NaOH:

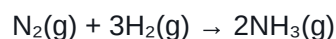
$$M(\text{NaOH}) = 23 + 16 + 1 = 40 \text{ g}\cdot\text{mol}^{-1}$$

Step 5 — Mass:

$$m = n M = 2,50 \times 10^{-3} \times 40 = 0,100 \text{ g}$$

Worked example 3 — limiting reagent and percentage yield

2,0 mol of nitrogen gas reacts with 4,0 mol of hydrogen gas:



(a) Identify the limiting reagent. (b) Calculate the mass of ammonia that should form. (c) If only 38,0 g of ammonia is obtained, calculate the percentage yield.

(a) H_2 needed for 2,0 mol $\text{N}_2 = 3 \times 2,0 = 6,0$ mol, but only 4,0 mol is available $\rightarrow \text{H}_2$ is the limiting reagent. Equivalently, from N_2 : $n(\text{NH}_3) = 2 \times 2,0 = 4,0$ mol; from H_2 : $n(\text{NH}_3) = (2/3) \times 4,0 = 2,67$ mol. The smallest value comes from H_2 .

$$(b) n(\text{NH}_3) = (2/3) \times 4,0 = 2,67 \text{ mol}$$

$$M(\text{NH}_3) = 14 + 3(1) = 17 \text{ g}\cdot\text{mol}^{-1}$$

$$\text{theoretical } m = n M = 2,67 \times 17 = 45,3 \text{ g}$$

$$(c) \% \text{ yield} = (\text{actual} / \text{theoretical}) \times 100 = (38,0 / 45,3) \times 100 = 83,9\%$$

Worked example 4 — molecular formula

A compound has an empirical formula of CH_2O and a molar mass of $180 \text{ g}\cdot\text{mol}^{-1}$. Determine its molecular formula.

Step 1 — Empirical formula mass:

$$M(\text{CH}_2\text{O}) = 12 + 2(1) + 16 = 30 \text{ g}\cdot\text{mol}^{-1}$$

Step 2 — Multiplier:

$$n = M_{\text{molecular}} / M_{\text{empirical}} = 180 / 30 = 6$$

Step 3 — Multiply every subscript by 6:

$$\text{Molecular formula} = \text{C}_6\text{H}_{12}\text{O}_6$$

Concentration, dilution and standard solutions

Concentration tells you how much solute is dissolved in a given volume of solution. A standard solution is one whose concentration is known accurately.

When a solution is diluted, the amount of solute (in moles) does not change, so:

$$c_1 V_1 = c_2 V_2$$

where the subscript 1 refers to before dilution and 2 to after dilution. Adding water increases the volume and lowers the concentration, but the number of moles of solute stays the same.

Example: 50,0 cm³ of 0,20 mol·dm⁻³ NaCl is diluted to 250 cm³; then $c_2 = c_1 V_1 / V_2 = (0,20 \times 0,0500) / 0,250 = 0,040 \text{ mol} \cdot \text{dm}^{-3}$.

Common mistakes

- Using grams instead of moles in a mole ratio. The ratio works only in moles.
- Forgetting to convert cm³ to dm³ before using $n = c V$.
- Not balancing the equation first, which gives the wrong ratio.
- Rounding too early. Keep extra figures and round only at the end.
- Confusing Avogadro's number (particles) with molar mass (grams). They are linked through n .
- Assuming the reactant with the larger mass is in excess — it is the number of moles and the ratio that matter.
- Confusing percentage yield (product obtained) with percentage purity (pure substance in a sample).

Quick summary / cheat sheet

- $n = m / M$; $N = n N_A$; $n = c V$; for a gas at STP, $V_m = 22,4 \text{ dm}^3 \cdot \text{mol}^{-1}$.
- $N_A = 6,022 \times 10^{23} \text{ mol}^{-1}$.
- $1 \text{ dm}^3 = 1\,000 \text{ cm}^3$.
- Balance first, then work in moles, then convert out.
- The limiting reagent gives the theoretical yield.
- % yield = actual / theoretical $\times 100$; % purity = pure / sample $\times 100$.
- Empirical formula = simplest ratio; molecular formula = whole-number multiple of it.

Practice questions

1. Calculate the number of moles in 8,8 g of carbon dioxide, CO₂. [2]
2. How many atoms are there in 0,25 mol of helium gas? [2]
3. Calculate the concentration of a solution made by dissolving 5,85 g of NaCl in enough water to make 500 cm³ of solution. [3]
4. 20,0 cm³ of 0,250 mol·dm⁻³ H₂SO₄ reacts completely with NaOH: $\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$. Calculate the mass of NaOH that reacts. [4]
5. A compound contains 40% carbon, 6,7% hydrogen and 53,3% oxygen by mass. Determine its empirical formula. [4]
6. In the reaction $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, 3,0 mol H₂ reacts with 1,0 mol O₂. Which is the limiting reagent, and what mass of water forms? [4]
7. The theoretical yield of a reaction is 120 g but only 96 g is obtained. Calculate the percentage yield. [2]

Answers

1. $M(\text{CO}_2) = 12 + 2(16) = 44 \text{ g} \cdot \text{mol}^{-1}$; $n = m / M = 8,8 / 44 = 0,20 \text{ mol}$.
1. $N = n N_A = 0,25 \times 6,022 \times 10^{23} = 1,51 \times 10^{23} \text{ atoms}$ (helium is monatomic).
1. $M(\text{NaCl}) = 23 + 35,5 = 58,5 \text{ g} \cdot \text{mol}^{-1}$; $n = 5,85 / 58,5 = 0,100 \text{ mol}$; $V = 500 \text{ cm}^3 = 0,500 \text{ dm}^3$; $c = n / V = 0,100 / 0,500 = 0,200 \text{ mol} \cdot \text{dm}^{-3}$.

1. $n(\text{H}_2\text{SO}_4) = c V = 0,250 \times 0,0200 = 5,00 \times 10^{-3} \text{ mol}$; ratio 1 : 2 $\rightarrow n(\text{NaOH}) = 1,00 \times 10^{-2} \text{ mol}$; $M(\text{NaOH}) = 40 \text{ g}\cdot\text{mol}^{-1}$; $m = 1,00 \times 10^{-2} \times 40 = 0,400 \text{ g}$.
1. C: $40 / 12 = 3,33$; H: $6,7 / 1 = 6,7$; O: $53,3 / 16 = 3,33$. Divide by 3,33: C = 1, H = 2, O = 1 $\rightarrow$ empirical formula CH_2O .
1. From H_2 : $n(\text{H}_2\text{O}) = 3,0 \text{ mol}$; from O_2 : $n(\text{H}_2\text{O}) = 2 \times 1,0 = 2,0 \text{ mol}$. The smaller value comes from O_2 , so O_2 is limiting. $n(\text{H}_2\text{O}) = 2,0 \text{ mol}$; $M(\text{H}_2\text{O}) = 18 \text{ g}\cdot\text{mol}^{-1}$; $m = 2,0 \times 18 = 36 \text{ g}$.
1. % yield = $(96 / 120) \times 100 = 80\%$.

Official EduRank Team notes. Always cross-check with your textbook and teacher.